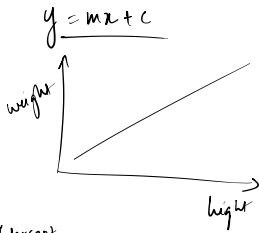


Regression

- ① Linear Regression is a statistical technique used to find the relationship between variables.

height (x) weight (y)



150	50
155	52
160	54
165	56
170	?

In line $y = mx + c$
slope $\downarrow$ intercept

x	y
height 1st	weight 1st
height 2nd	weight 2nd
height 3rd	weight 3rd

In Regression, $y = w \cdot x + b$
 $\downarrow$ $\downarrow$ $\downarrow$
 predicted weight input Bias
 label feature

$$y = w_1 x_1 + w_2 x_2 + b \rightarrow y = w_i x_i + b$$

- ① Column = Features

Rows = Data Points

y = Target [feature which we are going to predict]

Independent variables = x [features]

Dependent Variables = y [Target]

Linear Regression $\rightarrow$ Target should be continuous numerical value

There should be at least one independent variable on which our dependent variable [Target] depends.

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

Code Implementation :-

- ① Library $\rightarrow$ Scikit-Learn $\rightarrow$ model selection
 Matplotlib $\rightarrow$ data visualization
 pandas } data manipulation/transformation/cleaning
 numpy }

$$\begin{bmatrix} \text{row} & \text{columns} \\ \text{iloc} & [: , 2 :] \end{bmatrix} \text{ numerical feature}$$

iloc [: , :2]

$$\begin{bmatrix} : \\ : \end{bmatrix} \quad [: , -1] \text{ last}$$

$[: 2] \Rightarrow$ first two

- ② Steps $\rightarrow$ ① Data Collections

- ② Data Preprocessing

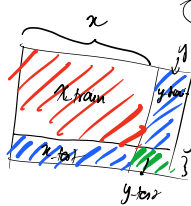
Data Cleaning, Data Transformation

EDA

Removing Duplicates

Handling missing value

Feature engineering



- ③ Define the features $[x]$ and Target $[y]$

- ④ Split the data into training set and testing set

x_{train}, x_{test}
 y_{train}, y_{test}

- ⑤ Train the model on training set
 $fit(x_{train}, y_{train})$
- ⑥ Predict the values from the model with the help of testing set as input.
 $\left[predict(x_{test}) \right] y_{pred}$
- ⑦ Compare the predicted value (y_{pred}) to the actual value (y_{test}) to find out the accuracy of the model.

